

Fast Photometric and Geometric Convergence for Online 3DGS

(Abstract) Mapping with 3D Gaussian Splatting (3DGS) benefits autonomous agents by enabling photorealistic rendering. However, in online mapping, the limited number of observations hinders rapid map convergence, while photometric loss alone can produce geometrically incorrect reconstructions, including floating artifacts in free space. We attribute these issues to two underlying causes: view selection tailored to camera pose estimation rather than map refinement, and insufficient control over which Gaussian parameters receive gradients from depth-based geometric losses. To address the former issue, we incorporate additional non-keyframe images into mapping and use Entropy-Regularized View Sampling to efficiently manage the resulting larger frame set by prioritizing frames that have received fewer updates. For geometric enhancement, we introduce a depth-guided Carve loss that directly updates opacity. It prevents optimization from circumventing free-space geometric constraints through compensatory changes in Gaussian positions or shapes. Together, these strategies enable the system to produce reliable geometry and appearance under the constraints of online environments.

1. Introduction

3D Gaussian Splatting (3DGS) is gaining attention for its photorealistic view synthesis and is widely used in vision-based autonomous agents [5]. However, its computationally intensive optimization is not suitable for online mapping, which is often required for active perception and action generation. Furthermore, poor geometric convergence leads to floaters in free space and poorly defined object surface boundaries. Recent studies use simultaneous localization and mapping (SLAM) systems to enable online learning, but inefficient view selection and insufficient consideration of geometry degrade mapping performance.

When integrating 3DGS into SLAM pipelines, using only SLAM keyframes for 3DGS optimization is insufficient due to the complexity of the scene representation. Thus, incorporating suitable frames into the training set while effectively allocating optimization steps across frames is important. We address this by incrementally adding non-keyframe images as mapping optimization progresses and introducing Entropy-Regularized View Sampling (ERVS) to prioritize frames that have received fewer updates, balancing new observations with continued refinement.

A limited number of viewpoints of the scene during online mapping poses another challenge: photometric optimization can converge to a local minimum that reproduces the observed images but yields inaccurate scene geometry. Depth-based losses provide geometric supervision, but their effect depends on which Gaussian parameters receive their gradients. Directly optimizing Gaussian positions leads to gradient discontinuities due to the discrete ordering in 3DGS rendering. Using shape as a constraint does not resolve the visibility of Gaussians in free space. Therefore, we introduce a causal, depth-guided Carve loss that directly suppresses Gaussian opacity in free space. By routing its gradients exclusively to opacity, we prevent changes in the positions or shapes of Gaussians in free space from compensating for this penalty.

Our contributions are summarized as follows:

- We formulate an online 3DGS mapping framework that

targets fast photometric and geometric convergence under limited computation.

- We introduce Optimization-Guided View Set Growth and ERVS to coordinate frame admission and update allocation with mapping progress.
- We design an opacity-directed causal Carve loss to suppress Gaussian contributions in observed free space through selective gradient routing.

2. Efficient Online Gaussian Mapping with Reliable Geometry

Our framework incrementally builds a Gaussian map from streaming RGB images and IMU measurements. A DROID-SLAM-based frontend [3] provides poses and depths to initialize and grow the Gaussian map. View Set Growth controls when new images are added to the training set, and View Sampling selects images from this set for each optimization update. The Carve loss adds depth-based geometric supervision to map optimization.

Optimization-Guided View Set Growth. Frames between SLAM keyframes can provide additional viewpoints and appearance information, but adding them to the training set leaves fewer optimization steps per image on average under a fixed budget. We therefore control the growth of the training view set by adding one available image after every κ optimization steps applied to the Gaussian map (Fig. 1(a)).

Entropy-Regularized View Sampling. Images added earlier to the training set have more opportunities for optimization. We distribute optimization steps more evenly across images while retaining randomness in image selection (Fig. 1(b)). We formulate this objective as

$$\min_{\mathbf{p} \in \Delta} \sum_i p_i \Delta_i \phi - \tau H(\mathbf{p}). \quad (1)$$

The first term favors less frequently selected images, while the entropy term encourages randomness in image selection. Here, $\mathbf{p} = (p_i)$ is the vector of image-selection probabilities, and Δ is the set of nonnegative vectors whose components sum

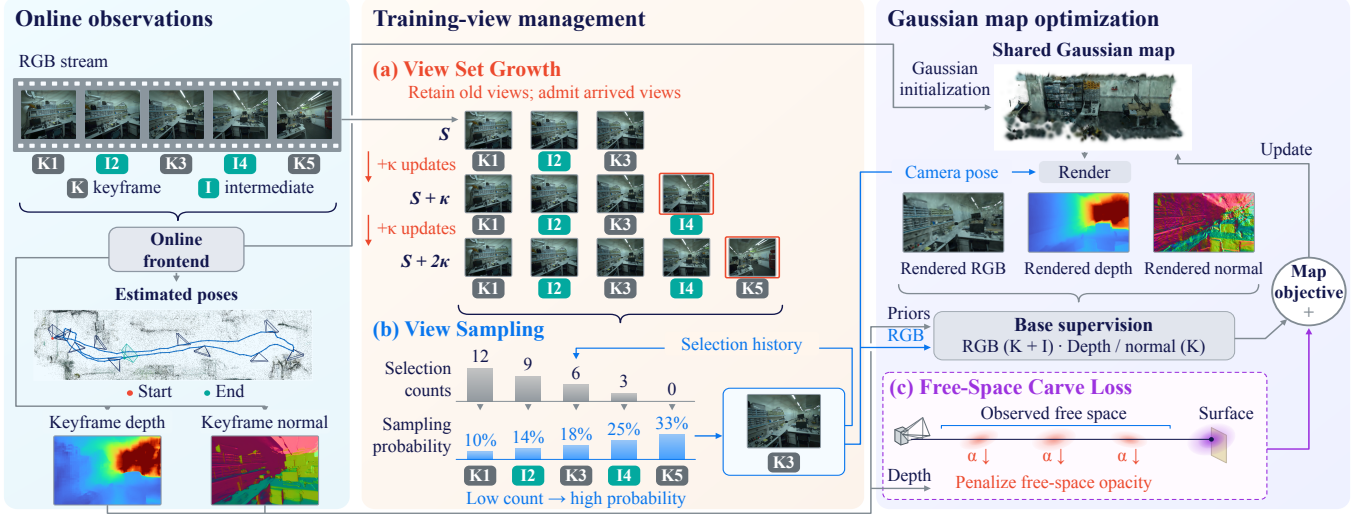


Fig. 1. Overview of the proposed online Gaussian mapping framework. (a) View Set Growth adds images to the training set based on the number of completed optimization steps. (b) View Sampling selects images from the training set for map optimization based on how often each image has been selected. (c) Carve uses available depth observations to suppress Gaussian opacity in observed free space.

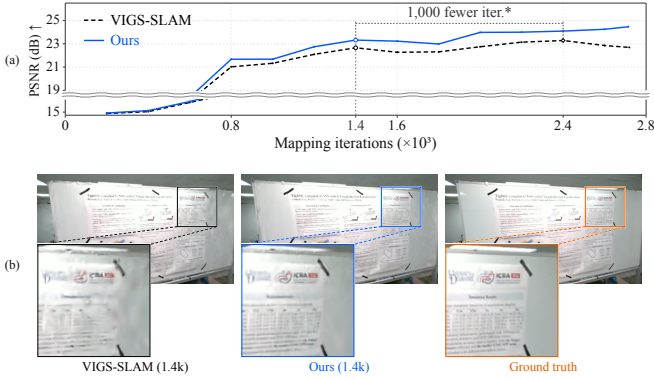


Fig. 2. Photometric convergence on RPNG [1] table_06. (a) PSNR over mapping iterations. *Our method first reaches the baseline’s highest PSNR with 1,000 fewer mapping iterations. (b) Renderings from VIGS-SLAM [4] (left) and our method (center) after 1,400 iterations, with ground truth (right). Enlargements highlight fine details.

to one. We measure imbalance by $\phi = \text{Var}(\mathbf{n}) / (2\bar{n})$, where $\mathbf{n} = (n_i)$ records cumulative selection counts and $\bar{n}$ is their mean. $\Delta_i\phi$ denotes the change in ϕ after selecting image i once more. The entropy is $H(\mathbf{p}) = -\sum_i p_i \log p_i$, with weight $\tau > 0$. **Causal Free-Space Carve Loss.** Depth observations provide evidence of free space between the camera and a surface. The Carve loss uses these observations to penalize Gaussian opacity in this free space, accounting for depth uncertainty and a margin around the surface (Fig. 1(c)). Its gradients act on opacity without updating Gaussian positions or rotations.

3. Results and Conclusion

We evaluate our method on two public RGB-IMU benchmarks, RPNG [1] and UTM [2], as well as on our own dataset collected using ARIA glasses. Under matched budgets for optimization steps, the average PSNR of our method on images excluded from training exceeds that of the base-

line [4] by 1.60, 0.52, and 2.45 dB, respectively. Our method requires fewer mapping iterations to achieve comparable rendering quality, enabling efficient online map refinement. After the same number of iterations, it produces clearer text and finer details than the baseline (Fig. 2).

We present an online Gaussian mapping system for efficient optimization of appearance and geometry, achieving faster photometric convergence and higher rendering quality under limited optimization budgets. Our system introduces view management to control training-set growth and allocate optimization steps across images, as well as a depth-guided Carve loss to suppress Gaussian opacity in free space. This approach provides a foundation for online mapping in robots and wearable augmented-reality systems that need reliable maps of their surroundings during operation, with limited computation and without waiting for offline refinement.

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